Measuring a Nation's Income

Primary Reference: Principles of Macroeconomics by Mankiw

Indraprastha Institute of Information Technology, Delhi

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Micro vs Macro

- *Microeconomics*: Individual households, firms and their interaction in the market.
- *Macroeconomics*: The economy as a whole. (Households, firms and markets simultaneously).

The Economy's Income and Expenditure

- Gross Domestic Product (GDP) measures two things:
  - The total income of everyone in the economy.
  - The total expenditure on the economy's output of goods and services.
- For an economy as a whole, income must equal expenditure.

The Economy's Income and Expenditure

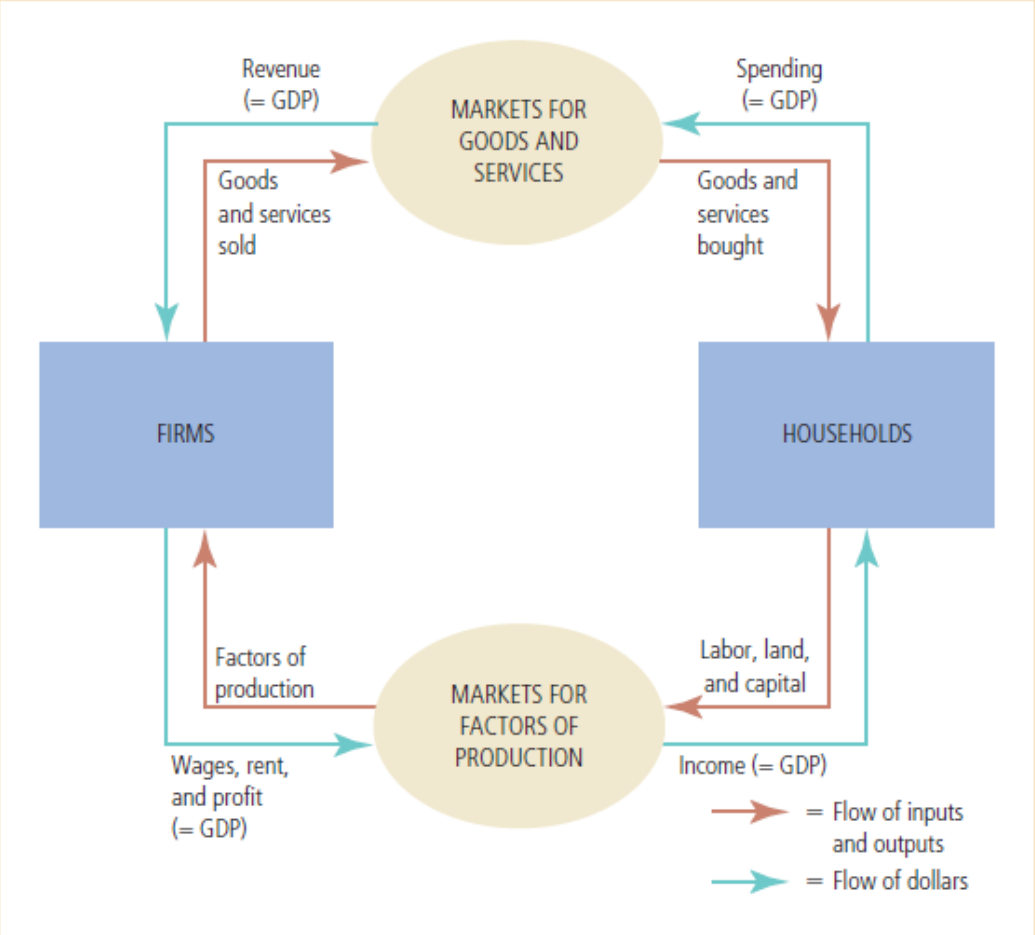

Figure 1

The Circular-Flow Diagram
Households buy goods and services from firms, and firms use their revenue from sales to pay wages to workers, rent to landowners, and profit to firm owners. GDP equals the total amount spent by households in the market for goods and services. It also equals the total wages, rent, and profit paid by firms in the markets for the factors of production.

**Source: Principles of Macroeconomics by N. Gregory Mankiw

The Measurement of GDP

- *GDP* is the market value of all final goods and services produced within a country in a given period of time.
  - Market value
  - of all
  - final
  - goods and services
  - produced
  - within a country
  - in a given period of time.

Components of GDP

$$Y = C + I + G + NX$$

- Y is GDP.
- C is consumption.
- I is investment.
- G is government purchases.
- NX is net exports.
  - Given by $(X - M)$.
  - X is exports.
  - M is imports.

Consumption

- Purchases made by households on goods and services.
- Housing is an exception.
- Examples include
  - Automobiles, appliances, food, clothing (goods).
  - Haircuts, medical care, education (services).

Investment

- Goods purchased to produce more goods in future.
- Sum of capital equipment, inventories and structures.
- What are inventories?
- Since GDP aims at measuring the production of an economy, inventories play a crucial role.

Government Purchases

- Spending by governments on goods and services.
- Can be local, state or central government.
- Different from transfer payments.
- Transfer payments alter household income, but do not reflect an economy's production.

Net Exports

- Exports minus imports
- Exports are the foreign purchases of domestically produced goods.
- Imports are the domestic purchases of foreign goods.

Calculating GDP

Table 1

GDP and Its Components

This table shows total GDP for the U.S. economy in 2009 and the breakdown of GDP among its four components. When reading this table, recall the identity $Y = C + I + G + NX$.

	Total (in billions of dollars)	Per Person (in dollars)	Percent of Total
Gross domestic product, Y	$14,259	$46,372	100%
Consumption, C	10,093	32,823	71
Investment, I	1,623	5,278	11
Government purchases, G	2,933	9,540	21
Net exports, NX	−390	−1,269	−3

Source: U.S. Department of Commerce. Parts may not sum to totals due to rounding.

**Source: Principles of Macroeconomics by N. Gregory Mankiw

Calculating GDP

GDP can be calculated three ways:

- Add up all spending on domestically produced final goods and services.
  - $Y = C + I + G + NX$.
- Add up the value added of all producers.
- Add up all income paid to factors of production.

Calculating GDP

2. Aggregate spending on domestically produced final goods and services = $21,500

	American Ore, Inc.	American Steel, Inc.	American Motors, Inc.	Total factor income
Value of sales	$4,200 (ore)	$9,000 (steel)	$21,500 (car)	
Intermediate goods	0	4,200 (iron ore)	9,000 (steel)	
Wages	2,000	3,700	10,000	$15,700
Interest payments	1,000	600	1,000	2,600
Rent	200	300	500	1,000
Profit	1,000	200	1,000	2,200
Total expenditure by firm	4,200	9,000	21,500	
Value added per firm = Value of sales – Cost of intermediate goods	4,200	4,800	12,500	

3. Total payments to factors = $21,500

1. Value of production of final goods and services, sum of value added = $21,500

**Source: Macroeconomics in Modules by Paul Krugman and Robin Wells

Real vs Nominal GDP

- When GDP increases, either,
  - Production has increased.
  - Prices have increased.
- It is important to separate these two effects.
- To do this, economists measure what is known as *real GDP*.
- Real GDP uses a base year's prices, while *nominal GDP* uses current year prices.

Real vs Nominal GDP

Table 2

Real and Nominal GDP

This table shows how to calculate real GDP, nominal GDP, and the GDP deflator for a hypothetical economy that produces only hot dogs and hamburgers.

Prices and Quantities

Year	Price of Hot Dogs	Quantity of Hot Dogs	Price of Hamburgers	Quantity of Hamburgers
2010	$1	100	$2	50
2011	$2	150	$3	100
2012	$3	200	$4	150

Calculating Nominal GDP

2010	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2011	($2 per hot dog × 150 hot dogs) + ($3 per hamburger × 100 hamburgers) = $600
2012	($3 per hot dog × 200 hot dogs) + ($4 per hamburger × 150 hamburgers) = $1,200

Calculating Real GDP (base year 2010)

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2011	($1 per hot dog × 150 hot dogs) + ($2 per hamburger × 100 hamburgers) = $350
2012	($1 per hot dog × 200 hot dogs) + ($2 per hamburger × 150 hamburgers) = $500

Calculating the GDP Deflator

2010	($200 / $200) × 100 = 100
2011	($600 / $350) × 100 = 171
2012	($1,200 / $500) × 100 = 240

**Source: Principles of Macroeconomics by N. Gregory Mankiw

The GDP Deflator

- The *GDP deflator* is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- The GDP deflator in the base year is always 100.

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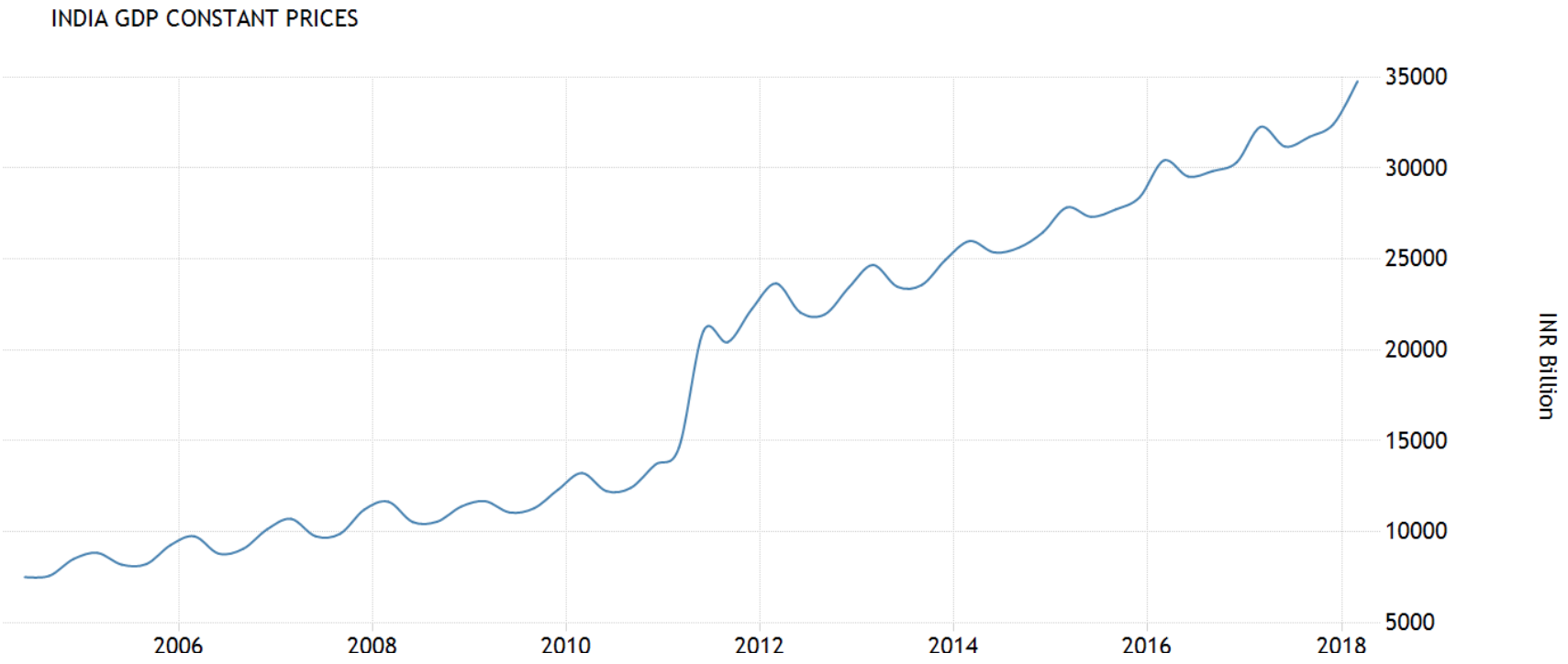
GDP Deflator

- GDP deflator can also be used to calculate inflation (though a more popularly accepted measure is Consumer Price Index or CPI).
- The inflation rate between two consecutive years is given as follows:

$$\text{Inflation rate in year 2} = \frac{\text{GDP deflator in year 2} - \text{GDP deflator in year 1}}{\text{GDP deflator in year 1}} \times 100$$

Real GDP over Recent History



**Source: Tradingeconomics.com | Ministry of Statistics and Program Implementation (MOSPI)

GDP as a Measure of Economic Well-Being

- Is it a good measure?
- Strengths?
- Weaknesses?

International Differences in GDP and Quality of Life

Table 3

GDP and the Quality of Life

The table shows GDP per person and three other measures of the quality of life for twelve major countries.

Country	Real GDP per Person (2007)	Life Expectancy	Adult Literacy (% of population)	Internet Usage (% of population)
United States	$45,592	79 years	99%	63%
Germany	34,401	80	99	45
Japan	33,632	83	99	67
Russia	14, 690	66	99	15
Mexico	14,104	76	93	18
Brazil	9,567	72	90	19
China	5,383	73	93	9
Indonesia	3,843	71	92	7
India	2,753	63	66	3
Pakistan	2,496	66	54	7
Nigeria	1,969	48	72	4
Bangladesh	1,241	66	54	0.3

Source: *Human Development Report 2009,* United Nations. Data on real GDP, life expectancy, and literacy are for 2007. Data on Internet use is for 2005.

**Source: Principles of Macroeconomics by N. Gregory Mankiw